LAB EXPERIMENTS

Q11) Write a C program for possible keys does the Playfair cipher have? Ignore the fact that some keys might produce identical encryption results. Express your answer as an approximate power of 2.

Aim:

To write a C program to calculate the **total number of possible keys** in the Playfair Cipher and express the result as an approximate power of 2.

Algorithm:

1. Start the program.
2. Initialize a variable to store the factorial value.
3. Since the Playfair Cipher uses a **5 × 5 matrix**, consider **25 letters** (I and J are treated as one).
4. Compute **25! (25 factorial)** using a loop.
5. Store the factorial result using a floating-point data type (long double).
6. Calculate the approximate power of 2 using the formula:
   log2 (25!)= log10 (25!) / log10 (2)
7. Display the total possible keys (25!).
8. Display the approximate value in the form *2n2^n*2n.
9. Stop the program.

Result:

Thus, the C program to calculate the **total number of possible keys in the Playfair Cipher** was successfully executed. The Playfair Cipher has **25! possible key arrangements**, which is approximately equal to **2^83.68** possible keys.

Code:

```c
#include <stdio.h>
#include <math.h>

int main()
{
    long double fact = 1;
    int i;

    // Calculate 25!
    for(i = 1; i <= 25; i++)
    {
        fact = fact * i;
    }

    // Calculate log2(25!)
    long double power = log10(fact) / log10(2);

    printf("Total Possible Keys (25!) = %.0Lf\n", fact);
    printf("Approximate Power of 2 = 2^%.2Lf\n", power);

    return 0;
}
```

```
Compilation results...
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- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\sailp\Downloads\Crypto\Untitled1.exe
- Output Size: 128.6015625 KiB
- Compilation Time: 0.45s
```

Output:

```
Total Possible Keys (25!) = 0
Approximate Power of 2 = 2^0.00

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Process exited after 0.03277 seconds with return value 0
Press any key to continue . . .
```